

Revision Notes for Class 12 Physics

Chapter 13 – Nuclei

Structure of an Atom:

- The atom consists of a central nucleus and extranuclear electrons.
- The nucleus is positively charged and contains protons and neutrons.
- Protons carry a positive charge, whereas neutrons are neutral.

Isotopes:

The atoms of an element, which constitute the same atomic number but different mass numbers are termed isotopes. Some examples are: ${}_8\text{O}^{16}$, ${}_8\text{O}^{17}$, ${}_8\text{O}^{18}$, ${}_{17}\text{O}^{35}$, ${}_{17}\text{Cl}^{37}$, ${}_{82}\text{Pb}^{206}$, ${}_{82}\text{Pb}^{207}$, ${}_{82}\text{Pb}^{208}$.

Isotones:

The atoms whose nuclei constitute the same number of neutrons are termed isotones.

Isobars:

The atoms which constitute the same mass numbers but different atomic numbers are termed isobars. Some examples are:

I. ${}_1\text{H}^3$ and ${}_2\text{He}^3$

II. ${}_3\text{Li}^7$ and ${}_4\text{Be}^7$

III. ${}_{28}\text{Ar}^{40}$ and ${}_{29}\text{Ca}^{40}$

IV. Ge^{76} and ${}_{34}\text{Se}^{76}$

Atomic Mass Unit:

Atomic mass unit (a.m.u) refers to a very small unit of mass and it is observed to be very convenient in nuclear physics.

It is defined as $1/12^{\text{th}}$ the mass of one ${}_6\text{C}^{12}$ atom.

With respect to Avogadro's hypothesis, number of atoms in 12 g of ${}_6\text{C}^{12}$ is taken equivalent to the Avogadro number i.e., 6.023×10^{23} .

Thus, the mass of one carbon atom (${}_6\text{C}^{12}$) is given by $\frac{12}{6.023 \times 10^{23}} = 1.992678 \times 10^{-26} \text{ kg}$

Further,

$$1 \text{ amu} = \frac{1}{12} \times 1.992678 \times 10^{-26} \text{ kg}$$

$$\Rightarrow 1 \text{ amu} = 1.660565 \times 10^{-27} \text{ kg}$$

Atomic Number (Z) and Mass Number (A):

- Atomic number (Z) represents the number of protons in the nucleus.
- Mass number (A) denotes the total number of protons and neutrons in the nucleus.
- Number of neutrons (N) can be calculated as $N = A - Z$.

Discovery of Neutron:

Scientist: James Chadwick discovered the neutron in 1932.

Nature of Neutron:

- Neutrons have no electrical charge (they are neutral).
- They have a mass similar to protons (about 1 atomic mass unit or amu).

Significance:

- The discovery helped explain why atomic nuclei, which have positively charged protons, don't repel each other.
- Neutrons contribute to the mass of the nucleus but do not affect its charge.

Size of the Nucleus:

1. **Concept of Nuclear Size:** The size of an atomic nucleus is very small compared to the size of the whole atom. It is typically on the order of femtometres (1 femtometre = 10^{-15} metres).

2. **Nuclear Radius Formula:** The radius of a nucleus can be estimated using the formula:

$$R = R_0 \times A^{1/3}$$

Where R is the radius of the nucleus, R_0 is a constant approximately equal to 1.2 to 1.3 fm, and A is the mass number of the nucleus.

3. **Example Calculation:** For a nucleus with mass number $A = 56$, the radius R can be calculated as:

$$R = 1.2 \times 56^{1/3} \text{ fm}$$

which gives an approximate radius.

4. Implication: This formula shows that the size of the nucleus increases with the mass number, but not in a linear way. It helps in understanding how nuclei compare in size as they get heavier.

Nuclear Stability:

- Stability of a nucleus depends on the ratio of protons to neutrons.
- For light nuclei ($Z < 20$), a roughly equal number of protons and neutrons ensures stability.
- For heavier nuclei, the optimal ratio of neutrons to protons increases.

Mass Energy and Nuclear Binding Energy

Mass-Energy

Concept: Mass-energy is the idea that mass can be converted into energy and vice versa. This is based on Einstein's famous equation: $E = mc^2$, where E is energy, m is mass, and c is the speed of light.

Explanation: In nuclear reactions, a small amount of mass is lost and converted into a large amount of energy. This principle explains why nuclear reactions release so much energy compared to chemical reactions.

Nuclear Binding Energy

Concept: Nuclear binding energy is the energy required to hold the nucleus of an atom together. It is the difference between the energy of the separate nucleons (protons and neutrons) and the energy of the nucleus when these nucleons are combined.

Explanation: When nucleons come together to form a nucleus, they lose some mass due to the binding energy. This energy is what keeps the nucleus stable.

Formula: The binding energy can be calculated using the mass defect. The mass defect is the difference between the mass of the fully assembled nucleus and the sum of the masses of its individual nucleons.

Formula: Binding Energy $E_B = \Delta m \times c^2$

Here, Δm is the mass defect, and c is the speed of light.

These concepts help explain how and why nuclear reactions produce so much energy and how stable atomic nuclei are formed.

Nuclear Forces:

- Nuclear forces are responsible for binding protons and neutrons together within the nucleus.
- These forces are strong but short-range in nature.
- They overcome the electrostatic repulsion between positively charged protons.

Radioactivity:

- Radioactive decay is the spontaneous disintegration of an unstable nucleus.
- Three common types of radioactive decay include alpha (α) decay, beta (β) decay, and gamma (γ) decay.
- Alpha decay emits an alpha particle (helium nucleus), beta decay releases beta particles (electrons or positrons), and gamma decay emits high-energy photons.

Radioactive Decay Law:

- Radioactive decay follows an exponential decay law: $N(t) = N_0 \times e^{(-\lambda t)}$.

- $N(t)$ represents the number of radioactive nuclei remaining at time t .
- N_0 is the initial number of radioactive nuclei, and λ is the decay constant.

Nuclear Energy: Nuclear energy is the energy released from changes in the nucleus of an atom. It comes from nuclear reactions, such as fission (splitting nuclei) and fusion (combining nuclei).

Energy Calculation: The energy released in nuclear reactions can be calculated using the formula: $E = m \cdot c^2$

where Δm is the mass defect (the difference in mass before and after the reaction) and c is the speed of the light (approximately) 3×10^8 m/s.

Applications: Nuclear energy is used in power plants to generate electricity and in medical treatments for things like cancer therapy. It plays a crucial role in providing a large amount of energy from small amounts of fuel.

Nuclear Fission and Fusion:

- Nuclear fission involves the splitting of a heavy nucleus into two lighter nuclei.
- Nuclear fusion is the process of combining two light nuclei to form a heavier nucleus.
- Both fission and fusion release an enormous amount of energy.

Controlled Thermonuclear Fusion: Controlled thermonuclear fusion is a process where atomic nuclei combine at high temperatures to release a large amount of energy. This is the same process that powers the sun and other stars.

How It Works: In fusion, light nuclei, like hydrogen isotopes (deuterium and tritium), fuse together to form a heavier nucleus, such as helium. This fusion releases energy due to the strong nuclear forces involved.

Conditions Required: To achieve controlled fusion, extremely high temperatures (millions of degrees) and pressures are needed to overcome the repulsive forces between the positively charged nuclei.

Formula: The energy released in a fusion reaction can be calculated using the mass-energy equivalence principle, given by Einstein's famous formula:

$$E = \Delta m \cdot c^2$$

where E is the energy released, Δm is the mass difference between the reactants and products, and c is the speed of light (approximately 3×10^8 metres/second).

Challenges: Creating and maintaining the conditions for controlled fusion is very challenging. Scientists are working on technologies like magnetic confinement (using devices like tokamaks) and inertial confinement (using lasers) to make fusion a practical and sustainable energy source.

Current Status: As of now, controlled thermonuclear fusion is still in the experimental stage, and researchers are striving to achieve a net positive energy output from fusion reactions.

Half-Life:

- Half-life $\left(T_{\frac{1}{2}}\right)$ is the time taken for half of the radioactive nuclei to decay.
- It is related to the decay constant as $\left(T_{\frac{1}{2}}\right) = \frac{\ln(2)}{\lambda}$.

Nuclear Reactors:

- Nuclear reactors utilise controlled fission reactions to generate electricity.
- They consist of a nuclear fuel, moderators, control rods, and a coolant.

- Control rods absorb excess neutrons to regulate the reaction.

Atoms Nucleus

- The atomic nucleus is a small, dense region located at the center of an atom.
- Ernest Rutherford discovered the atomic nucleus in 1911 through the Geiger-Marsden gold foil experiment conducted in 1909.
- The nucleus is positively charged and consists of protons and neutrons.
- Surrounding the nucleus is a cloud of negatively charged electrons.
- The nucleus and electron cloud are bound together by electrostatic force.
- The majority of the mass of an atom is concentrated in the nucleus.
- The electron cloud contributes very little to the overall mass of the atom.
- Protons and neutrons in the nucleus are held together by the nuclear force.

Forces

- Nuclei are bound together by the residual strong force, also known as the nuclear force.
- The residual strong force binds quarks together to form protons and neutrons.
- The nuclear force is highly attractive at typical nucleon separation distances, overcoming the electromagnetic repulsion between protons.
- Nuclei exist because the attractive nuclear force overcomes the electromagnetic repulsion.
- The residual strong force has a limited range and decays quickly with distance.

- Only nuclei smaller than a certain size can be completely stable due to the limited range of the residual strong force.
- The largest completely stable nucleus is lead-208, which contains 208 nucleons (126 neutrons and 82 protons).
- Nuclei larger than lead-208 are unstable and tend to have shorter lifetimes.
- Bismuth-209 is stable to beta decay and has the longest half-life to alpha decay among all known isotopes.
- The residual strong force is effective over a very short range, typically only a few femtometers.
- The residual strong force causes an attraction between any pair of nucleons.

Advances and Challenges

- Ongoing research efforts are focused on nuclei with up to eight nucleons to measure various properties of light.
- The simplest nuclei provide the basis for quantitative comparisons between experimental and theoretical maps of their global and short-range structure.
- Light nuclei are ideal for probing the microscopic aspects of nuclear structure, especially those related to gluons and quarks.
- Light nuclei play important roles in astrophysics, elementary particle physics, and energy production.
- The majority of matter in the visible universe exists in the form of light nuclei.
- Reactions between light nuclei primarily govern the nuclear physics of conventional stars like the Sun and the Big Bang.
- Free neutrons are unstable and undergo radioactive decay.

- Helium-3 (^3He) and Deuterium (^2H) are useful surrogates for neutron targets in comparative measurements of the internal structure of neutrons to protons.
- A detailed understanding of the structure of these nuclei is necessary for interpreting experimental results.
- Electron scattering is a direct method for probing the structure of nuclei.
- The nucleus, located at the core of an atom, is a quantal many-body system governed by the strong interaction.
- Neutrons and protons, as the most stable hadrons, compose the nucleus, just like gluons and quarks compose hadrons.
- The binding of nucleons together in nuclei through the strong force is a fundamental question related to the existence of the universe.
- Mutual interactions between nucleons shortly after the Big Bang led to the formation of light nuclei.